

The Reduced Exact Coordinate MILP Implemented in `gurobi_coordinate_solver.py`

Abstract

This document states only the `ReducedExactCoordinateMILP` constructed by `gurobi_coordinate_solver.py`. This is the model selected by `configs/gurobi_coordinate_default.json`. Each physical box has one integer local coordinate and one orientation, while pallet assignment remains binary. Pairwise direction variables prevent boxes assigned to the same pallet from overlapping. Optional support constraints use exact horizontal intersection areas. The implemented objective minimizes the number of used pallets. All sets, parameters, variables, auxiliary linearizations, and their sources in the input or configuration are described below.

1 Modeling idea

Every candidate pallet has the same dimensions and coordinate system. A box is assigned to exactly one pallet, so it needs only one local position and one orientation. It does not need a separate copy of its coordinates for every candidate pallet. Geometry is therefore indexed by boxes and box pairs; pallet indices occur only in assignment, pallet activation, capacity, and the constraints that activate non-overlap for boxes sharing a pallet.

The word *exact* refers to this reduction: replacing pallet-indexed copies of identical local geometry by one state per box does not approximate the integer-grid packing problem. Exactness remains relative to the configured grid, allowed orientations, maximum candidate-pallet count, and active constraints. Section 8 records an important caveat concerning two optional pallet-label restrictions enabled together in the supplied configuration.

2 Input data and preprocessing

2.1 Sources

The model is built from two sources:

1. An MCPP/PL-100 JSON instance supplies the pallet and an `items` or `boxes` array.
2. A JSON configuration supplies grid resolution, candidate-pallet and item limits, rotation and support modes, symmetry options, Gurobi parameters, and the type of area auxiliaries. The supplied default configuration selects `model_variant = reduced_exact` and `area_auxiliary_type = integer`.

The parser accepts several dimension-field schemas. After normalization, let $(L^{\text{mm}}, W^{\text{mm}}, H^{\text{mm}})$ denote the physical pallet dimensions and $(\ell_i^{\text{mm}}, w_i^{\text{mm}}, h_i^{\text{mm}})$ the physical dimensions of box i .

2.2 Grid conversion

Let $g \in \mathbb{Z}_{>0}$ be the grid size in millimetres. It is read from `grid_mm`; if this value is the string `input`, it is read from `meta.grid_mm` in the instance, with 50 mm as the parser fallback. Pallet

dimensions are rounded down:

$$(L, W, H) := \left(\left\lfloor \frac{L^{\text{mm}}}{g} \right\rfloor, \left\lfloor \frac{W^{\text{mm}}}{g} \right\rfloor, \left\lfloor \frac{H^{\text{mm}}}{g} \right\rfloor \right). \quad (1)$$

The unrotated dimensions of box i are rounded upward:

$$(\ell_i, w_i, h_i) := \left(\max \left\{ 1, \left\lceil \frac{\ell_i^{\text{mm}}}{g} \right\rceil \right\}, \max \left\{ 1, \left\lceil \frac{w_i^{\text{mm}}}{g} \right\rceil \right\}, \max \left\{ 1, \left\lceil \frac{h_i^{\text{mm}}}{g} \right\rceil \right\} \right). \quad (2)$$

This is conservative: the usable pallet is not enlarged and the boxes are not shrunk. A feasible integer-grid packing therefore remains within the physical pallet boundary, subject to the small numerical tolerances used while parsing.

2.3 Sets and indices

Let n be the number of records in the instance. Model construction requires $1 \leq n \leq n^{\text{max}}$, where n^{max} is the configuration value `max_items`. This is a preprocessing guard, not a MILP constraint. Define

$$\mathcal{I} := \{0, \dots, n-1\}, \quad i, j \in \mathcal{I}, \quad (3)$$

$$\mathcal{P} := \{0, \dots, P^{\text{max}} - 1\}, \quad p \in \mathcal{P}, \quad (4)$$

$$\mathcal{E} := \{(i, j) \in \mathcal{I}^2 : i < j\}, \quad (5)$$

$$\vec{\mathcal{E}} := \{(i, j) \in \mathcal{I}^2 : i \neq j\}. \quad (6)$$

The candidate-pallet count P^{max} is determined before model construction: an explicit command-line `-max-pallets` value takes precedence. If it is absent, the code takes the maximum of the configured value and an available instance recommendation. That recommendation is read from `meta.bounds.ub_pallets_heuristic`, or otherwise from the last entry of `meta.bounds.certified_optimum_in`. Candidate pallets are physically identical; their labels have no physical meaning.

For each $i \in \mathcal{I}$, preprocessing first generates distinct orientation triples. The configuration value `rotation_mode` has the following meaning:

none	only (ℓ_i, w_i, h_i) ;
yaw	(ℓ_i, w_i, h_i) and (w_i, ℓ_i, h_i) ;
metadata	the same yaw-only behavior in the current code;
six	all distinct permutations of the three dimensions.

If `upright_only` is true, only the two yaw orientations are allowed even in **six** mode. Duplicate triples caused by equal side lengths are removed. The reduced model then removes every orientation that does not fit inside (L, W, H) . The resulting set and its fixed dimensions are

$$\mathcal{O}_i := \{0, \dots, m_i - 1\}, \quad (d_{io}^x, d_{io}^y, d_{io}^z) \in \mathbb{Z}_{>0}^3, \text{quado} \in \mathcal{O}_i. \quad (7)$$

Every retained triple satisfies $d_{io}^x \leq L$, $d_{io}^y \leq W$, and $d_{io}^z \leq H$. An item with no retained orientation is rejected before the MILP is built. The current exception path contains a minor implementation bug: it refers to `item_id`, whereas the data class field is `id`.

2.4 Parameters used by the MILP

Symbol	Source	Meaning
L, W, H	instance and Eq. (1)	pallet extents in integer grid cells; also axis-specific big- M constants.
$P^{\max}$	CLI, config, and possibly instance bounds	number of identical candidate pallets.
d_{io}^a	instance dimensions, grid, rotation mode	extent of box i on axis $a \in \{x, y, z\}$ in orientation o .
v_i	$\ell_i w_i h_i$	conservative box volume in grid cells; invariant under rotation.
V	LWH	pallet volume in grid cells.
K^{vol}	$\max\{1, \lceil \sum_i v_i / V \rceil\}$	volume lower bound on the number of pallets.
$\bar{q}_{ij}^x$	$\min\{\max_o d_{io}^x, \max_o d_{jo}^x\}$	pair-specific upper bound on horizontal overlap length in x .
$\bar{q}_{ij}^y$	$\min\{\max_o d_{io}^y, \max_o d_{jo}^y\}$	pair-specific upper bound on horizontal overlap length in y .
$A^{\max}$	LW	safe global upper bound for contact-gating a footprint area.
$\bar{A}_j$	$\max_o d_{jo}^x d_{jo}^y$	maximum possible base area of box j .
$\tilde{\alpha} = r/s$	support configuration	rationalized required support fraction, with denominator at most 1000.
M^{loc}	$L(W+1)(H+1) + W(H+1) + H$	big- M for position ordering of identical boxes.

If `support.mode = full`, then $\tilde{\alpha} = 1$. In `fraction` mode, the code converts the configured floating-point value α to a decimal string and applies a denominator-limited rational approximation:

$$\tilde{\alpha} := \text{limitDenominator}_{1000}(\text{Fraction}(\text{str}(\alpha))) = \frac{r}{s}. \quad (8)$$

Thus the MILP enforces $\tilde{\alpha}$, which need not equal an arbitrary configured α . The supplied value $0.75 = 3/4$ is represented exactly. The post-solve geometry audit instead checks the original floating-point value.

2.5 Parsed attributes not used as physical constraints

The parser reads or defaults the following additional item fields: ID, SKU, original millimetre dimensions, weight, `volume_dm3`, family, food, chemical, fragile, upright-only, and retrieval priority. ID and SKU default from the record index; weight defaults to 1 kg; family and the food/chemical flags may be inferred from `group`; the Boolean handling flags default to false; priority falls back from `retrieval_priority` to `priority` and then to 1.

The optimization volume is $v_i = \ell_i w_i h_i$, not the parsed `volume_dm3`. Weight, food/chemical status, fragility, family, and retrieval priority do not impose payload, compatibility, accessibility, or handling constraints in this solver. Some of them are used only in the identical-item key in Section 8; the fields are otherwise retained for output and future extensions.

3 Decision variables

3.1 Core variables

Variable	Index	Domain	Meaning
u_p	$p \in \mathcal{P}$	$\{0, 1\}$	1 iff candidate pallet p is used.
a_{ip}	$i \in \mathcal{I}, p \in \mathcal{P}$	$\{0, 1\}$	1 iff box i is assigned to pallet p .
r_{io}	$i \in \mathcal{I}, o \in \mathcal{O}_i$	$\{0, 1\}$	1 iff orientation o is selected for box i .
x_i, y_i, z_i	$i \in \mathcal{I}$	respectively $\mathbb{Z} \cap [0, L]$, $\mathbb{Z} \cap [0, W]$, $\mathbb{Z} \cap [0, H]$	local lower-left-bottom coordinates.
$\bar{x}_i, \bar{y}_i, \bar{z}_i$	$i \in \mathcal{I}$	same respec- tive integer bounds	right, front, and top coordinates.
λ_{ij}^x	$(i, j) \in \vec{\mathcal{E}}$	$\{0, 1\}$	1 iff box i ends at or before box j starts on x .
λ_{ij}^y	$(i, j) \in \vec{\mathcal{E}}$	$\{0, 1\}$	analogous relation on y (“in front of”).
λ_{ij}^z	$(i, j) \in \vec{\mathcal{E}}$	$\{0, 1\}$	analogous relation on z (“below”).
f_j	$j \in \mathcal{I}$	$\{0, 1\}$	1 iff box j rests on the pallet floor.
c_{ij}	$(i, j) \in \vec{\mathcal{E}}$	$\{0, 1\}$	1 when box i is selected as a direct supporter of box j .
S_{ij}	$(i, j) \in \vec{\mathcal{E}}$	$\mathbb{Z}_{\geq 0}$ by sup- plied config	footprint area of the pair counted as support from i to j .

The support variables f_j, c_{ij}, S_{ij} exist only in the modes described in Section 7. The reduced model has no separate base-area variable. Its oriented base area is the linear expression

$$B_i := \sum_{o \in \mathcal{O}_i} d_{io}^x d_{io}^y r_{io}. \quad (9)$$

3.2 Auxiliaries for horizontal overlap area

In **fraction** and **full** support modes, every unordered pair $(i, j) \in \mathcal{E}$ receives the following variables. These are always created by the reduced model. The configuration option `overlap_formulation` is not read by this formulation.

Variable	Index	Domain	Meaning
m_{ij}^x	$(i, j) \in \mathcal{E}$	$\mathbb{Z} \cap [0, L]$	$\min\{\bar{x}_i, \bar{x}_j\}$.
M_{ij}^x	$(i, j) \in \mathcal{E}$	$\mathbb{Z} \cap [0, L]$	$\max\{x_i, x_j\}$.
δ_{ij}^x	$(i, j) \in \mathcal{E}$	$\mathbb{Z} \cap [-L, L]$	signed difference $m_{ij}^x - M_{ij}^x$.
q_{ij}^x	$(i, j) \in \mathcal{E}$	$\mathbb{Z} \cap [0, L]$	exact positive overlap length $\max\{0, \delta_{ij}^x\}$.
m_{ij}^y	$(i, j) \in \mathcal{E}$	$\mathbb{Z} \cap [0, W]$	$\min\{\bar{y}_i, \bar{y}_j\}$.
M_{ij}^y	$(i, j) \in \mathcal{E}$	$\mathbb{Z} \cap [0, W]$	$\max\{y_i, y_j\}$.
δ_{ij}^y	$(i, j) \in \mathcal{E}$	$\mathbb{Z} \cap [-W, W]$	signed difference $m_{ij}^y - M_{ij}^y$.
q_{ij}^y	$(i, j) \in \mathcal{E}$	$\mathbb{Z} \cap [0, W]$	exact positive overlap length $\max\{0, \delta_{ij}^y\}$.
β_{ijk}	$(i, j) \in \mathcal{E}$, bit k	$\{0, 1\}$	binary expansion of one chosen overlap length.
t_{ijk}	same	$\mathbb{Z}_{\geq 0}$ by supplied config	product of bit β_{ijk} and the other overlap length.

Variable	Index	Domain	Meaning
A_{ij}	$(i, j) \in \mathcal{E}$	$\mathbb{Z}_{\geq 0}$ by supplied config	exact footprint-intersection area $q_{ij}^x q_{ij}^y$.

If `area_auxiliary_type = continuous`, t_{ijk} , A_{ij} , and S_{ij} are continuous instead. The supplied configuration declares them integer. The reduced class also uses integer as its fallback if the option is omitted.

4 Objective

The only optimization objective inserted into Gurobi is

$$\min \sum_{p \in \mathcal{P}} u_p. \quad (10)$$

It minimizes the number of used pallets. No weight, center-of-mass, accessibility, compatibility, or height objective is present.

5 Assignment, orientation, and containment

Each box is assigned to exactly one pallet and receives exactly one retained orientation:

$$\sum_{p \in \mathcal{P}} a_{ip} = 1 \quad \forall i \in \mathcal{I}, \quad (11)$$

$$\sum_{o \in \mathcal{O}_i} r_{io} = 1 \quad \forall i \in \mathcal{I}. \quad (12)$$

Assignment activates a pallet, and a pallet may not be marked used without a box:

$$a_{ip} \leq u_p \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, \quad (13)$$

$$u_p \leq \sum_{i \in \mathcal{I}} a_{ip} \quad \forall p \in \mathcal{P}. \quad (14)$$

Define the oriented dimensions as linear expressions

$$D_i^a := \sum_{o \in \mathcal{O}_i} d_{io}^a r_{io} \quad a \in \{x, y, z\}. \quad (15)$$

The endpoints are

$$\bar{x}_i = x_i + D_i^x, \quad \bar{x}_i \leq L, \quad (16)$$

$$\bar{y}_i = y_i + D_i^y, \quad \bar{y}_i \leq W, \quad (17)$$

$$\bar{z}_i = z_i + D_i^z, \quad \bar{z}_i \leq H \quad \forall i \in \mathcal{I}. \quad (18)$$

Together with nonnegative coordinates, these constraints keep every oriented box inside its assigned pallet's local coordinate system.

The following volume inequality is a valid strengthening for each pallet:

$$\sum_{i \in \mathcal{I}} v_i a_{ip} \leq V u_p \quad \forall p \in \mathcal{P}. \quad (19)$$

If $K^{\text{vol}} > P^{\text{max}}$, the solver rejects the instance before completing the model. Otherwise, the reduced implementation fixes

$$u_p = 1 \quad p = 0, \dots, K^{\text{vol}} - 1. \quad (20)$$

It does not insert a separate, redundant aggregate row $\sum_p u_p \geq K^{\text{vol}}$.

6 Pairwise non-overlap

The direction variables describe exact interval separation. For every ordered pair $(i, j) \in \vec{\mathcal{E}}$,

$$\bar{x}_i \leq x_j + L(1 - \lambda_{ij}^x), \quad x_j \leq \bar{x}_i - 1 + L\lambda_{ij}^x, \quad (21)$$

$$\bar{y}_i \leq y_j + W(1 - \lambda_{ij}^y), \quad y_j \leq \bar{y}_i - 1 + W\lambda_{ij}^y, \quad (22)$$

$$\bar{z}_i \leq z_j + H(1 - \lambda_{ij}^z), \quad z_j \leq \bar{z}_i - 1 + H\lambda_{ij}^z. \quad (23)$$

Because the coordinates are integer, these paired inequalities imply, for example,

$$\lambda_{ij}^x = 1 \iff \bar{x}_i \leq x_j. \quad (24)$$

If the relation is false, $x_j \leq \bar{x}_i - 1$, so the two x intervals overlap strictly or box j lies completely to the left. Touching faces count as separation, as required for non-overlap.

For each unordered pair, at least one of the six separating directions must hold whenever the boxes share any candidate pallet:

$$\begin{aligned} & \lambda_{ij}^x + \lambda_{ji}^x + \lambda_{ij}^y + \lambda_{ji}^y + \lambda_{ij}^z + \lambda_{ji}^z \\ & \geq a_{ip} + a_{jp} - 1 \quad \forall (i, j) \in \mathcal{E}, p \in \mathcal{P}. \end{aligned} \quad (25)$$

If i and j are assigned to different pallets, every right-hand side is at most zero and their local coordinates need not be separated. If they share pallet p , the right-hand side is one, which prevents volumetric overlap.

7 Support constraints

The configuration selects one of four modes: **off**, **direct**, **fraction**, or **full**. In **off** mode, no variables or constraints from this section are created.

7.1 Floor and direct-contact logic

For every box j , the floor indicator is exact:

$$z_j \leq H(1 - f_j), \quad z_j \geq 1 - f_j. \quad (26)$$

Thus $f_j = 1$ iff $z_j = 0$.

For every ordered pair $(i, j) \in \vec{\mathcal{E}}$, contact may be selected only when the two boxes have the same pallet assignment. The code imposes, for every pallet,

$$c_{ij} \leq 1 - a_{ip} + a_{jp}, \quad c_{ij} \leq 1 + a_{ip} - a_{jp} \quad \forall p \in \mathcal{P}. \quad (27)$$

Because each assignment vector contains exactly one 1, these inequalities force $c_{ij} = 0$ when the boxes are on different pallets. Exact vertical contact is activated by

$$z_j - \bar{z}_i \leq H(1 - c_{ij}), \quad \bar{z}_i - z_j \leq H(1 - c_{ij}). \quad (28)$$

Hence $c_{ij} = 1$ implies $\bar{z}_i = z_j$: box i touches the bottom of box j .

7.2 Direct support mode

In direct mode, binary overlap indicators $o_{ij}^x, o_{ij}^y \in \{0, 1\}$ are created for $(i, j) \in \mathcal{E}$:

$$o_{ij}^x \leq 1 - \lambda_{ij}^x, \quad o_{ij}^x \leq 1 - \lambda_{ji}^x, \quad o_{ij}^x \geq 1 - \lambda_{ij}^x - \lambda_{ji}^x, \quad (29)$$

$$o_{ij}^y \leq 1 - \lambda_{ij}^y, \quad o_{ij}^y \leq 1 - \lambda_{ji}^y, \quad o_{ij}^y \geq 1 - \lambda_{ij}^y - \lambda_{ji}^y. \quad (30)$$

These are exact indicators of strictly positive overlap on the two horizontal axes. Contact is gated by

$$c_{ij} \leq o_{\min\{i,j\},\max\{i,j\}}^x, \quad c_{ij} \leq o_{\min\{i,j\},\max\{i,j\}}^y. \quad (31)$$

Every box must then be on the floor or have at least one direct supporter with positive footprint overlap:

$$f_j + \sum_{i \in \mathcal{I} \setminus \{j\}} c_{ij} \geq 1 \quad \forall j \in \mathcal{I}. \quad (32)$$

The size of the positive overlap does not matter in this mode.

7.3 Fractional and full support modes

In fraction and full modes, the reduced formulation computes the exact horizontal rectangle intersection for every $(i, j) \in \mathcal{E}$. Gurobi general constraints implement

$$\begin{aligned} m_{ij}^x &= \min\{\bar{x}_i, \bar{x}_j\}, & M_{ij}^x &= \max\{x_i, x_j\}, \\ \delta_{ij}^x &= m_{ij}^x - M_{ij}^x, & q_{ij}^x &= \max\{0, \delta_{ij}^x\}, \end{aligned} \quad (33)$$

$$\begin{aligned} m_{ij}^y &= \min\{\bar{y}_i, \bar{y}_j\}, & M_{ij}^y &= \max\{y_i, y_j\}, \\ \delta_{ij}^y &= m_{ij}^y - M_{ij}^y, & q_{ij}^y &= \max\{0, \delta_{ij}^y\}. \end{aligned} \quad (34)$$

Therefore, the exact footprint-intersection area is

$$A_{ij} = q_{ij}^x q_{ij}^y. \quad (35)$$

The product is linearized exactly using a binary expansion. The code compares the bit counts required by $\bar{q}_{ij}^x$ and $\bar{q}_{ij}^y$ and expands the shorter representation. Let q_{ij}^b denote the selected overlap length, q_{ij}^c the other one, $\bar{q}_{ij}^b$ and $\bar{q}_{ij}^c$ their bounds, and

$$K_{ij} := \max\left\{1, \left\lceil \log_2(\bar{q}_{ij}^b + 1) \right\rceil\right\}. \quad (36)$$

The inserted linearization is

$$q_{ij}^b = \sum_{k=0}^{K_{ij}-1} 2^k \beta_{ijk}, \quad (37)$$

$$t_{ijk} \leq q_{ij}^c, \quad (38)$$

$$t_{ijk} \leq \bar{q}_{ij}^c \beta_{ijk}, \quad (39)$$

$$t_{ijk} \geq q_{ij}^c - \bar{q}_{ij}^c (1 - \beta_{ijk}), \quad (40)$$

$$A_{ij} = \sum_{k=0}^{K_{ij}-1} 2^k t_{ijk}. \quad (41)$$

With $t_{ijk} \geq 0$, Eqs. (38)–(40) give $t_{ijk} = \beta_{ijk} q_{ij}^c$ exactly.

Only an area belonging to an activated direct contact may count as support:

$$S_{ij} \leq A_{\min\{i,j\},\max\{i,j\}}, \quad (42)$$

$$S_{ij} \leq A^{\max} c_{ij}, \quad (43)$$

$$S_{ij} \geq A_{\min\{i,j\},\max\{i,j\}} - A^{\max} (1 - c_{ij}). \quad (44)$$

Thus $S_{ij} = A_{\min\{i,j\},\max\{i,j\}}$ when $c_{ij} = 1$ and $S_{ij} = 0$ when $c_{ij} = 0$.

With $\tilde{\alpha} = r/s$, every box must satisfy

$$s \sum_{i \in \mathcal{I} \setminus \{j\}} S_{ij} + r \bar{A}_j f_j \geq r B_j \quad \forall j \in \mathcal{I}. \quad (45)$$

For a raised box, $f_j = 0$, so the selected direct-support areas must cover at least the rationalized fraction r/s of its oriented base area. For a floor box, $f_j = 1$ and $B_j \leq \bar{A}_j$, so the row is automatically satisfied. In full mode, $r/s = 1$.

8 Pallet-label and identical-item restrictions

The model always orders candidate-pallet use as a prefix:

$$u_p \geq u_{p+1} \quad p = 0, \dots, P^{\max} - 2. \quad (46)$$

This is valid because pallet labels are interchangeable.

If `order_pallet_loads` is enabled, the code also imposes

$$\sum_{i \in \mathcal{I}} v_i a_{ip} \geq \sum_{i \in \mathcal{I}} v_i a_{i,p+1} \quad p = 0, \dots, P^{\max} - 2. \quad (47)$$

If `fix_first_item` is enabled, it imposes

$$a_{0,0} = 1. \quad (48)$$

Important interaction. Each of Eqs. (47) and (48) can select a valid pallet labeling in isolation, but their simultaneous use is not guaranteed to preserve a representative of every feasible packing. For example, suppose a $10 \times 1 \times 1$ pallet must hold boxes of lengths 6, 5, and 5, with the length-6 box indexed first. The only two-pallet arrangement puts the length-6 box alone and the two length-5 boxes together. Fixing the first box to pallet 0 gives loads 6 and 10, whereas load ordering requires $6 \geq 10$. The supplied default configuration currently enables both options. Consequently, with those settings the overall configured model can exclude a feasible packing even though the reduced coordinate geometry itself is exact. For an exact pallet-label reduction, at least one option should be disabled or the two conditions should be replaced by one compatible canonicalization.

If `order_identical_items` is enabled, boxes are grouped by the exact key

$$(\text{SKU}, (\ell_i, w_i, h_i), \text{weight}, \text{family}, \text{food}, \text{chemical}, \text{fragile}, \text{uprightOnly}, \text{priority}). \quad (49)$$

For every consecutive index pair (i, j) in a group,

$$\sum_{p \in \mathcal{P}} p a_{ip} \leq \sum_{p \in \mathcal{P}} p a_{jp}. \quad (50)$$

Define the injective integer encoding of a grid point

$$\phi(x, y, z) := x(W + 1)(H + 1) + y(H + 1) + z. \quad (51)$$

When both boxes are assigned to pallet p , their local positions are ordered:

$$\phi(x_i, y_i, z_i) \leq \phi(x_j, y_j, z_j) + M^{\text{loc}}(2 - a_{ip} - a_{jp}) \quad \forall p \in \mathcal{P}. \quad (52)$$

These identical-item constraints remove permutations of records that the implementation regards as interchangeable; they do not add a physical packing requirement.

9 Model summary and limitations

The reduced MILP chooses one pallet, one retained orientation, and one integer coordinate for every physical box. It minimizes used pallets while enforcing pallet activation, conservative grid containment, volume capacity, pairwise non-overlap, the configured support rule, and enabled label/order restrictions.

The formulation is discrete: it does not search arbitrary continuous coordinates, and conservative rounding may exclude packings that require sub-grid dimensions or positions. It also does not enforce payload weight, center of mass, density, food–chemical separation, fragile-box restrictions, family compatibility, retrieval accessibility, or a height-balancing objective. Those attributes should not be inferred from the generated solution merely because they are present in the input and output records.